Modeling Transcription Start Site Activity With 5' RNA Sequencing at Single-Cell Resolution

A thesis presented in partial fulfilment of the requirements for the degree of Bachelor of Science

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Abstract

The regulation of gene expression requires complex interactions between cis-regulatory elements, such as promoters and enhancers, and trans-regulatory factors like transcription factors (TFs). In unison, these elements maintain precise control of gene expression, and are essential for the components of cellular differentiation.

Advances in high-throughput sequencing, such as RNA-sequencing (RNA-seq) and ATAC-sequencing (ATAC-seq), have largely expanded our understanding of transcriptional activity and chromatin accessibility. RNA-seq measures gene expression profiles, while ATAC-seq pinpoints open regions of chromatin. Although insightful, neither sequencing method links transcriptional outcomes to specific regulatory elements, which leaves gaps in our understanding of these elements' role in transcription regulation.

Using genomic sequence features and transcriptional data, machine learning models have addressed this limitation. Transformer-based architectures have demonstrated the ability to predict transcriptional activity and chromatin accessibility by piling sequencing reads near a genomic locus and using the underlying DNA sequencing or quantitative cell state as model input. In this study, we adapt an existing transformer model to enhancer-specific datasets.

We leverage a novel dataset of non-overlapping, bidirectional enhancer sequences in this thesis, to explore whether an existing transformer framework linked with our 5' RNA sequencing data can capture transcription initiation. Through this approach, we discovered we could successfully predict general RNA coverage, as well as transcription start site usage. In doing this, we have taken a key step toward modeling regulatory elements such as enhancers

and promoters.

Acknowledgements

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Chapter 1

Introduction

This chapter provides a thesis introduction and describes the objectives of the paper.

1.1 Research Objective and Motivation

In this thesis, we adapt a transformer model named 'Scooby', (developed by the Gagneur lab from the Technical University of Munich)[1] with our own enhancer-sequence centered dataset. In harnessing this novel dataset, we aim to improve upon and advance Scooby's previous accomplishments in single-cell genetic profile prediction using RNA-only data.

In short, Scooby predicts RNA coverage at a cell-state specific level by training on every read mapped in RNA-seq[1] data, regardless of their origin on the transcript. However, gene transcription can initiate from different positions, depending on the cell type. This variation cannot be detected with 3'[2] RNA-seq data, as it only captures the ends of transcripts. In order to observe these differences, we need to analyze the actual start sites of RNA molecules, which 5'[3] RNA-seq data makes possible. With that, in this study we adapt Scooby to use 5' data to investigate whether transformer-based models can predict start site activity in a cell-state-specific manner.

[1]See Appenidix A.1 for at length definition
[2]See Appenidix A.2 for at length definition
[3]See Appenidix A.3 for at length definition

Biologically, this thesis was motivated by the need to identify sequence-based sources of variation in transcription start site usage across single cells. Since alternate transcription start site usage is a known lineage-dependent outcome of gene regulation, capturing these patterns from the underlying DNA sequence offers insight into transcriptional control. We aim to use this model in future work to test variability in transcription start site usage in a model system. Organoids offer an ideal future data source for this, as their cell lineages are traceable, allowing for easier validation on sequence-driven predictions of transcription start site usage.

To start, we will reproduce the original results of Scooby, utilizing its original data set to validate model implementation. In addition, this will give us a baseline of prediction performance on cell-state-specific RNA coverage. Next, we will curate a novel dataset focused on 5' enhancer sequences, representing non-overlapping, bidirectional transcriptional activity. This 5' region of DNA includes promoters and transcription start sites, critical elements of transcription initiation. Generally speaking, this region serves as a central hub for regulatory activity. Focusing on this region allows for the analysis of core regulatory elements responsible for gene expression, like transcription factor (TF) binding sites and motifs. In addition, this concentrated scope removes the presence of added complexity from long-range enhancer interactions. Using 5'-centered data, we will first determine whether the model can still predict cell-state specific RNA coverage, now using signals concentrated around transcription initiation sites in training. Finally, we will test whether the model can go *beyond* general coverage prediction to learn patterns of start site activity, enabling cell-specific modeling of transcription initiation. Accomplishing this would extend the capabilities of the original Scooby model, in turn contributing new insights into regulatory genomics while increasing the scope of genomic modeling.

Chapter 2

Foundations of Gene Regulation

This chapter provides an overview of the fundamental concepts of gene regulation and modern profiling techniques, offering a foundation for understanding the advancements in gene modeling through machine learning.

2.1 Regulatory Mechanisms

The central dogma of molecular biology describes the flow of genetic information from DNA to RNA to protein. The activity of cis-regulatory elements like enhancers and promoters tightly regulates this process. More specifically, they encode transcription factor binding domains, along with the activity of trans-regulatory elements like transcription factors, which promote polymerase binding. Promoters are located near the genes they regulate, marking the starting point of transcription, while enhancers can act over long distances, attracting TFs to modify chromatin accessibility and increase transcription rates. As traditional molecular biology creates a strong distinction between the two functional entities, modern findings have pointed towards their activity being more context-dependent and overlapping. These insights challenge our traditional idea on the relationship between regulatory elements, encouraging further investigation on their interconnected, but dynamic functions. In addition to cis-regulatory mechanisms, trans-regulatory factors play a critical role in shaping tran-

scriptional outcomes. These are the proteins that bind to DNA motifs in enhancers and promoters, through their interaction with cis-elements and mediation of broader regulatory networks. This interplay between cis and trans-regulatory mechanisms is fundamental in orchestrating gene expression. Cis-elements provide sequence specific targets for TFs to bind to, in turn activating or repressing transcription. The complexity of gene expression is constructed from these systems, and allows cells to adapt and differentiate in many biological contexts.

2.2 Sequencing Methods

Modern methods such as RNA sequencing and ATAC sequencing[1] have revolutionized our understanding of cell life cycles and genetic expression. Manifold embeddings of high-dimensional RNA-seq data provide a snapshot of gene expression by quantifying RNA molecules of a cell into a mathematical vector space. This offers insights into the transcriptional activity of cells between points in their life cycles. The integration of ATAC-seq aids this representation through the identification of open regions of chromatin in DNA, in turn revealing where TFs and various regulatory proteins can bind and interact with the genome. With these methods, scientists can map the genetic activity of cells throughout their various states, in turn giving them an idea of how cells develop during processes like differentiation. Lineage tracing[2] adds another dimension to these methods. This is a process of physically 'marking' cells early in their development to track their genetic progression. This tactic adds the dimension of time to our understanding of how cells are developing.

While insightful, little can be revealed about regulatory elements and their impact on gene expression, with only sequencing methods. Approaches like RNA-seq measure the transcriptional outcomes of cells, but lack the ability to attribute these outcomes to specific cis-regulatory elements such as enhancers and promoters. In a similar vein, lineage tracing

[1] See Appendix A.4 for at length definition
[2] See Appenidix A.5 for at length definition

offers a perspective on cell fate and differentiation, however it does not provide insight into regulatory sequences' influence on these processes. Machine learning models have rapidly narrowed this gap in understanding, as they are able to utilize DNA sequence data to predict transcriptional activity and chromatin accessibility. With the help of genomic sequence features (like enhancer and promoter sequences) and high-resolution output data (like ATAC-seq), these models architect a way to decode complex relationships between cis-regulatory elements and gene expression. With these developments, we are offered unprecedented insights into the 'black-box-like' regulatory framework of the genome. The next chapter will delve into some of the models that broke this barrier in modeling gene expression, which form the basis for the research performed in this thesis.

Chapter 3

A Comprehensive Review of Machine Learning Applications in Gene Expression Profiling

This chapter will highlight some of the impactful models which shape the way machine learning and predicting gene expression intertwine. These models not only motivate this research topic, but provided a strong foundation from which we could build upon for our own adapted models.

3.1 Machine Learning Principles

Machine learning models analyze masses of data in the process of uncovering intricate patterns, enabling them to make predictions with high accuracy. As the field of machine learning has rapidly advanced in recent years, a variety of model architectures have gained popularity, driving significant progress across numerous domains. In this thesis, we focus on transformer models; while convolutional neural networks (CNNs) have long been a foundational architecture in machine learning, transformers represent a newer and more complex framework that has revolutionized the field in many ways. The most notable advancement in using

transformers is ChatGPT, a model that exemplifies their potential.

3.1.1 Transformer Models

Transformers are a revolutionary type of machine learning architecture that have (quite literally) transformed the field of artificial intelligence. Originally developed by researchers at Google, transformers are primarily built upon the multi-head attention mechanism, which allows the model to process and contextualize sequential data. The architecture begins by converting input data into numerical representations called tokens, which are then mapped into vector representations. From here, the embeddings are processed through multiple layers of the architecture–each layer refines the representation by focusing on relationships within the input sequence. This is accomplished with self-attention[1], the mechanism that truly defines transformers.

The core of transformer architectures lies in their self-attention mechanism, specifically the multi-head attention process. In standard self-attention, a single set of query, key, and value projections is used to compute attention scores (level of importance of the data) and weighted outputs for each position in the input sequence. While effective, this limits the model to learning only one set of attention patterns. By using multiple attention heads, multi-head attention has many different mechanisms each learning different patterns of interactions within the data. The heads operate in unison to capture relationships between inputs and produce a richer representation of the input sequence. With this, transformers excel in capturing local *and* global dependencies–this makes them particularly effective on jobs involving text, DNA sequences, and other types of sequential data.

[1]See Appendix A.6 for at length definition

3.2 Borzoi Model — Predicting RNA-seq coverage from DNA Sequences

3.2.1 Overview

Borzoi[2] is a state-of-the-art model designed to predict cell-specific RNA-seq coverage directly from DNA sequences. This convolutional neural network incorporates self-attention mechanisms, enabling it to capture both local and global dependencies within large sections of DNA sequences. To reiterate, self-attention is an architecture in machine learning models which allows the model to dynamically focus on various parts of the input sequence, and identify regions and patterns most relevant for making predictions. Borzoi uses these features to effectively model the complex relationships between genomic sequences and RNA expression levels, making it a powerful tool for understanding transcriptional regulation.

Borzoi's ability to process large amounts of training data is key, as it enhances its predictive accuracy and ability to generalize. The Scooby model, which we will cover next, builds upon Borzoi's foundation, adapting it to predict both RNA-seq coverage and chromatin accessibility for specific cell types.

Borzoi's reliance on large genomic regions as input allows it to consider the broader context of regulatory elements, such as enhancers and promoters, which influence RNA expression. This comprehensive approach ensures that Borzoi captures the full range of sequence features relevant to transcriptional activity. The inclusion of self-attention further strengthens the model by enabling it to dynamically focus on key regions of the DNA sequence, adapting its analysis to the specific patterns relevant for each prediction.

3.2.2 Model Architecture

Let us take a closer look at the architecture behind the Borzoi model.

Borzoi is heavily based on the Enformer model[2], a preceding model in the field. The architecture begins with convolutional and downsampling layers (which use max pooling) which extract local sequence patterns and reduce the input sequence length. This new length corresponds to 128 base pairs (bp), which allows for more generalized processing. From here, the embeddings pass through a self-attention block with relative positional encodings, helping Borzoi capture long-range interactions and relationships across the entire sequence, despite its downsampling.

Following self-attention, the output is up-sampled through a process which doubles its resolution. This up-sampled output is created using a feature map from the convolutional layers via U-Net[3] connections. The U-Net architecture in effect 'saves' much of the fine-grained features lost during downsampling, in order to reapply them at this step. Additional convolutional layers are applied to refine the predictions further, in the end producing RNA-seq coverage at a final resolution of 32 bp bins. To avoid less accurate predictions on the sequence boundaries (boundaries are susceptible to lack of filter context through the nature of convolutional layers), the model focuses its loss computation on the central base pairs of the sequence by cropping the edges.

Borzoi's training data is comprised of a diverse set of RNA-seq data. It includes 867 human and 278 mouse RNA-seq coverage tracks with input sequences as long as 524 thousand base pairs (kb). Such an extensive training set ensures Borzoi's robustness and helps it achieve a high accuracy. To optimize performance, Borzoi incorporates two loss functions. First, a Poisson loss on the total observed vs. predicted coverage, and second, a multinomial loss on the normalized coverage distribution across the sequence length. The heavier weighted multinomial loss ensures the model accurately captures the relative distribution of RNA-seq coverage. By comparison, the Poisson loss enforces accuracy with absolute count values. The paper notes the combination of loss functions significantly enhanced prediction performance.

[2]See Appendix A.7 for at length definition
[3]See Appendix A.8 for at length definition

Borzoi introduces simplifications and refinements over the Enformer architecture to better suit RNA-seq predictions. Its ability to integrate local patterns, long-range dependencies, and precise coverage predictions makes it a powerful tool for understanding transcriptional activity. The self-attention layers, U-Net architecture, and joint loss function create a unique model that stands as a strong basis for RNA-seq coverage prediction.

3.3 Scooby Model — Modeling genomic profiles from DNA sequences at single-cell resolution

3.3.1 Overview

Scooby is an advanced model that builds upon the Borzoi model previously discussed. As mentioned, Borzoi's transformer architecture can capture intricate patterns in genomic data, making it a solid foundation for further developments. Scooby builds upon this by freezing the pre-trained weights of Borzoi and applying Low-Rank Adaptation (LoRA)[4] (a fine-tuning technique that allows the model to specialize in specific cell types). This targeted fine-tuning ensures that Scooby retains the generalizable features learned by Borzoi, while adapting to the unique characteristics of individual cell types.

Scooby additionally incorporates a decoder mechanism, expanding its capabilities beyond those of its predecessor (encoders and decoders are algorithms that convert data from one format to another). In Scooby, the decoder takes the encoded representations of single-cell RNA-seq (scRNA-seq) and scATAC-seq data and converts them into predictions of gene expression counts and chromatin accessibility. In capturing both transcriptional activity and the regulatory landscape, Scooby's dual functionality offers a holistic understanding of cellular states.

As stated, Scooby predicts counts at specific genomic positions for each gene, precisely

[4]See Appendix A.9 for at length definition

modeling gene expression patterns. Second, it predicts chromatin accessibility across the genome, providing insights into the regulatory elements that influence gene activity. By integrating these complementary datasets, Scooby not only advances the field of single-cell analysis but also provides a powerful tool for studying the interplay between gene expression and chromatin structure.

Scooby's design demonstrates the growing importance of multi-modal approaches in computational biology. Its ability to synthesize information from separate data types represents a significant step forward in our understanding of complex cellular systems. By combining the strengths of the Borzoi model with innovative fine-tuning and decoding strategies, Scooby establishes itself as a state-of-the-art framework for single-cell genomics research.

3.3.2 Model Architecture

The Scooby model specifically leverages scRNA-seq and scATAC-seq data from the multiome hematopoiesis dataset used in the NeurIPS 2021 challenge. To prevent information leakage and artificial performance between cell embeddings and prediction models, genes and peaks (regions of open chromatin) present in the test and validation sets were identified and excluded prior to embedding. Further, genes and peaks appearing in fewer than 1% of all cells were filtered out, ensuring a more robust dataset. The scRNA-seq and scATAC-seq data were unified using the MultiVI model from the scvi python package, providing an efficient embedding for modeling, summarizing the intricate and noisy raw sequence data.

To process the data, the SnapATAC[5] python package was employed to convert the multimodal data into Anndata format, compiling the coverage into 32 base-pair windows. Following Borzoi's approach, Scooby modeled raw fragment counts using a Poisson distribution. For ATAC profiles, the output was scaled by a factor of 0.05 to align it with the scale of RNA coverage tracks.

Scooby integrates LoRA modules into the architecture, applying them with a rank of 8 to

[5]See Appendix A.10 for at length defintion

all convolutional, self-attention, and multi-layer perceptron matrices within the transformer layers. At a high level, this LoRA selectively modifies only specific components of a pre-trained model. This increases flexibility in the single-cell context of Scooby, while allowing for efficient fine-tuning at a relatively low computational cost. This was implemented using the Peft[6] python module. Next, these LoRA modules were merged back into the original model weights to streamline the architecture, despite them being trained separately. In addition, a trainable layer with GELU nonlinearity was added on top of Borzoi's penultimate layer. This added layer allows Scooby to adapt Borzoi's structure in the single-cell context while leaving room for further fine-tuning.

The output of Scooby's refined architecture is passed through a cell-specific decoder. Each decoder is generated for every cell based on its corresponding embedding. A small multi-layer perceptron is included to achieve this task, as it takes the cell embedding as input, and outputs a vector to parameterize the decoder filter. This filter produces the final predicted profiles for each cell. For both RNA-seq and ATAC-seq predictions, the model outputs a matrix that predicts the counts for each segment of the sequence.

[6]See Appendix A.11 for at length definition

Chapter 4

Methods

This chapter will provide an overview of the methods we used for the research in the paper. It will include model adaptation breakdowns, an analysis of the data we curated for the research, and our methodology for validation and testing.

4.1 Data

The data pipeline employed in this thesis was developed by Matthew Karikomi and has been integral to the project's success. Built to handle multi-batch, single-cell RNA-seq datasets, this pipeline ensures consistent and reproducible preprocessing, enabling the downstream modeling and analyses presented here.

4.1.1 Data Collection

For this study, we utilized data from the GENCODE Human Release 47 dataset, a high-quality resource for genome annotation which provides detailed information on both protein-coding and non-coding RNA genes. This dataset includes files essential for the genomic and transcriptomic analysis completed in this paper. This includes gene annotation files (in GTF format) which define gene structural organization. Additionally, transcript sequence

files (in FASTA format) are used, which contain sequences for protein-coding genes, long non-coding RNAs, and the full genomic DNA sequence. In addition, functional categorization of the genes is provided by gene classification files, which aides in transcriptome-wide analyses. With these resources, we can accurately align sequencing reads and map transcript sequences for expression analysis for 5' end RNA sequencing.

The 5' RNA-seq data used in this study is processed and loaded from pre-aligned fragment files, which store sequencing read alignments mapped to TSS regions, enabling the extraction of promoter-proximal transcriptional activity. This data was collected across ten batches of human fetal gut tissue (FCA-801 and FCA-945). It provides a diverse set of cell types and transcriptional states for our model training. Locus annotation files further define genomic regions of interest, ensuring the model is trained on the most biologically relevant features, such as promoters and other regulatory elements. By leveraging GENCODE annotations, we can establish that the extracted sequences align with well-characterized genomic features, improving the accuracy of downstream analyses. These processed 5' RNA-seq reads serve as our novel training data for our transformer model, enabling them to predict RNA expression patterns based on promoter activity. As for evaluation, we used a separate dataset derived from 3' RNA sequencing to evaluate model performance across different transcriptomic profiling techniques.

4.1.2 Data Preprocessing

The preprocessing pipeline outlined below processes 5' RNA-seq data to ultimately ensure the alignment of sequencing reads, extraction of TSS information, and accurate feature quantification. This pipeline automates key tasks such as genome indexing, read alignment, soft-clipped read extraction, and quantification, in turn optimizing the data for downstream analysis.

The preprocessing begins with the generation of a genomic index that enables the efficient alignment of sequencing reads to the reference genome. This step valuably applies a

filtering which retains high-confidence cells while removing low-quality ones, adding a layer of robustness to the data. Following this alignment, the pipeline performs soft-clipped read extraction, which is critical for the identification of TSS. This is achieved using a soft-clipping approach that isolates partially aligned reads, allowing for the detection of transcriptional activity near the promoter, which is a key characteristic of 5' end RNA sequencing.

In simplified terms, aligning sequencing reads to the reference genome allows us to trace each RNA fragment back to its original location in the DNA, helping to identify active genes and understand how gene expression is regulated. Since RNA is transcribed from DNA, mapping these fragments provides key insights into transcriptional activity. The process is further refined through the soft-clipping process as it preserves small RNA fragments that do not perfectly align to the reference genome.

4.1.3 Data Analysis

Following preprocessing, our data pipeline integrates and processes gene expression data, preparing it for downstream modeling. During preprocessing, sequencing reads were aligned to the genome, and gene expression matrices were generated for individual batches, capturing transcript counts for each cell. These matrices, inherited from preprocessing, represent raw expression data before normalization and feature extraction. Since each batch may have been processed separately, variations in sequencing depth, technical noise, and batch effects require the combination of these matrices into a single unified dataset to create a consistent framework for analysis. With that, loading and merging these gene expression matrices is the first step in this section of the pipeline. This will ensure that barcode and feature metadata are correctly processed. Addtionally, we leveraged the Harmony [3] architecture, an algorithm that projects cells into cell-type specific groups.

Once the dataset is structured, we use the aligned sequencing reads to extract TSS activity data, a key step in capturing promoter activity in 5' RNA-seq. We applied bias correction and compute background peaks to prove our analysis is not skewed. These corrections allow

for a more accurate representation of transcriptional activity and prevent biases that could distort downstream modeling results.

We then match motifs by comparing sequencing data to JASPAR[1] TF motifs. This allows us to infer potential TF binding sites as well as regulatory mechanisms. After this, we normalize and log-transform gene expression counts to make the data comparable across all cells, again ensuring robustness for our downstream analysis. To visualize high-dimensional transcriptomic data, we perform Principal Component Analysis (PCA)[2] and Uniform Manifold Approximation and Projection (UMAP)[3]. This reduces the complexity of the data set while preserving meaningful biological variation.

We then identify highly variable genes which aid in distinguishing different cell states. Using Leiden clustering[4], we segment the dataset into distinct cell populations at multiple resolutions, helping to uncover meaningful biological subtypes. To identify genes that are uniquely active in specific clusters, a differential gene expression analysis was done using Wilcoxon[5] and logistic regression tests.

These methods of structuring, normalizing, and clustering our dataset ensure that our modeling process is grounded in *biologically meaningful* features. The resulting clusters and differentially expressed genes provide interpretable inputs for our predictive models, improving their ability to capture key regulatory patterns and transcriptional activity from 5' RNA-seq data. Leveraging these structured datasets to expand Scooby's predictive power forms the center of this research.

To conclude, our goals in this paper revolve around the idea that leveraging these structured datasets will enhance our model's ability to predict and generalize across diverse biological conditions.

[1]See Appendix A.12 for at length definition
[2]See Appendix A.13 for at length definition
[3]See Appendix A.14 for at length definition
[4]See Appendix A.15 for at length definition
[5]See Appendix A.16 for at length definition

4.2 Scooby Model Adaptation

Unlike the original Scooby model designed for multiome data (RNA + ATAC-seq), we adapted Scooby to function as an RNA-only model to improve its predictive power for scRNA-seq data. Since our 5' dataset exclusively contains RNA expression data, using an RNA-only version of Scooby eliminates dependencies on chromatin accessibility signals, which allows the model to focus only on transcriptomic regulatory patterns.

To achieve this, we restricted model input to RNA signals. In this, we adjusted the model architecture to operate on just two tracks (RNA+ and RNA-), as opposed to the original three-track setup (RNA+, RNA-, and ATAC). Consequently, LoRA fine-tuning in the RNA-only model is applied exclusively to RNA-related layers, ensuring that the model learns RNA-specific regulatory interactions without being influenced by chromatin accessibility features.

Furthermore, in the multiome version of Scooby, LoRA fine-tunes interactions between RNA and ATAC, capturing relationships between gene expression and chromatin accessibility. However, in our RNA-only model, LoRA is limited to fine-tuning gene expression prediction, which results in different convergence behavior during training. Additionally, by removing the ATAC-related layers, these layers no longer have pre-trained weights and are therefore randomly initialized during the LoRA process. As a result, the fine-tuning process was longer as more time was required for the model to achieve optimal performance in an RNA-only setting.

We will be adapting Scooby into two distinct models in this paper, soon to be introduced as Scooby-5'-cov and Scooby-3'-tss. These models will both follow the same general structure outlined above, and will be trained on 5' RNA-seq data. The core difference of the two comes in what they predict; Scooby-5'-cov models coverage across all mapped reads at known loci, while Scooby-5'-tss only models transcription start site coverage. Scooby-5'-cov can be seen as a proof of concept for our 5' data experiments (to confirm coverage can be predicted with start-site centric data), while Scooby-5'-tss is the true extension of Scooby's capabilities.

4.2.1 Reproducing Results

To validate the findings of the Gagneur lab and obtain a baseline for RNA coverage prediction, we attempted to reproduce their published results using the baseline johahi/borzoi-replicate-0 pretrained model from HuggingFace. The reported baseline evaluation of their pre-trained model, which worked with both RNA and ATAC data, yielded a mean Pearson R score of 0.32 across cell types, establishing a benchmark for performance.

Further, we fine-tune the model using the NeurIPS RNA-only dataset. However, the initial results were significantly worse, with mean Pearson R scores dropping below 0.05 (shown in figure 4.3), indicating a mismatch between the dataset and the pre-trained model's learned representations. From figure 4.1, we can see the model was essentially learning to predict 0 every time, and from figure 4.2, we see there was little-to-no correlation between our predictions and highly expressed cells (which are typically easier to predict).

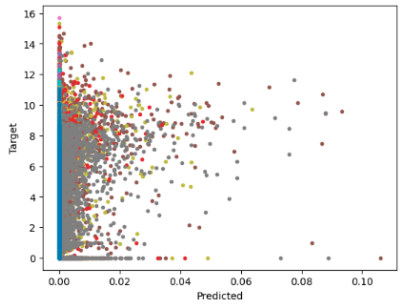

Figure 4.1: Scatter plot of Predictions vs. Targets

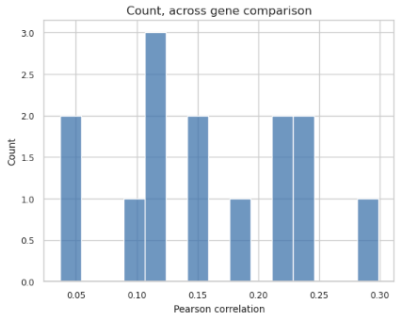

Figure 4.2: Histogram of Pearson R Scores for highl expression genes

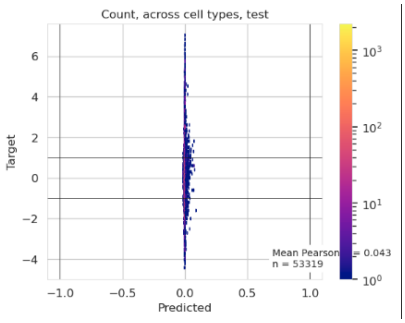

Figure 4.3: Cell Type-Level Predictions

Figure 4.4: Evaluation of Scooby-3'-cov Performance Across Metrics–Pre-adjusted

Through numerous modification attempts, such as adjusting weight decay, loss scaling, sequence clipping, and normalization techniques, we ultimately matched the original **0.32** Pearson R score across cell types after fine-tuning, which can be seen in figure 4.7. Figure 4.5 shows the weak, positive correlation obtained by Scooby between predicted vs. actual signal per locus, across all cells. Additionally, for highly expressed genes, shown in the histogram in figure 4.6 (genes with expression levels over 1), the original Scooby model produced a

median Pearson R score of **0.77**.

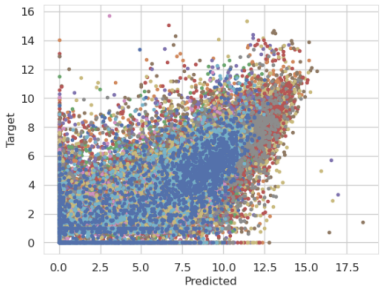

Figure 4.5: Scatter plot of Predictions vs. Targets

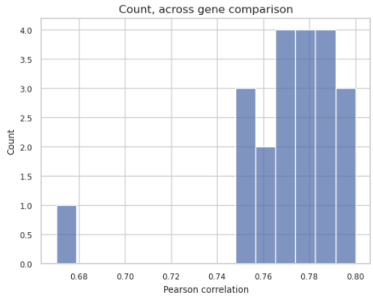

Figure 4.6: Histogram of Pearson R Scores for highl expression genes

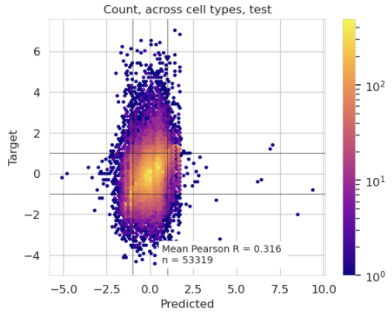

Figure 4.7: Cell Type-Level Predictions

Figure 4.8: Evaluation of Scooby-3'-cov Performance Across Metrics

These Pearson R scores acted as a strong baseline for performance of RNA coverage prediction using only RNA sequencing data. While a detailed layout of our modifications to Scooby's original architecture can be found in Appendix B, our main adjustments are summarized below:

First, we increased the target smoothing scaling factor c_{upscale} (See Appendix B — Convolutional Smoothing of RNA Targets) to improve training stability on sparse 5' RNA-seq data. Second, we applied z-score normalization to the smoothed target vectors, ensuring the model received numerically stable input during training (See Appendix B — Target Normalization). These changes were constant in all other models we trained in this thesis. Moving forward, our baseline Scooby model (whose results are shown in figures 4.5-4.7) will be referred to as **Scooby-3'-cov**.

4.2.2 Inputs

The original Scooby model was trained with paired multiomic datasets which incorporated both DNA accessibility and RNA expression to model regulatory activity. Its inputs included a reference genome, locus BED files for training regions, paired chromatin accessibility and RNA fragment files, and cell annotations with PCA embeddings and neighborhood graphs.

These components enabled the model to associate genomic sequences with functional output in different cell states.

In contrast, our fine-tuned Scooby models with 5' RNA-seq data use fragment files that reflect 5' TSS coverage. This is enabled by soft-clipping pre-processing (explained further in Section 4.1.2), which ensures that coverage is anchored to transcription start sites. The new training pipeline also retains the reference genome and annotated loci, however the targets for supervision represent start-site usage rather than full-transcript coverage. Additionally, curated cell annotations and PCA embeddings are used to define neighborhood relationships across cell types, just as in the original model. Overall, the training setup mirrors the original, but with data reflecting 5' read structure to enable TSS prediction.

We developed two models from this foundation, both using the same 5' RNA-seq input. One model incorporates a BED file of known genomic loci during training and is optimized to predict general RNA coverage across the transcript. Moving forward, this model will be referenced as **Scooby-5'-cov**. The second model is trained in the absence of positional information and focuses on capturing transcription start site activity. Moving forward, this model will be referenced as **Scooby-5'-tss**.

4.3 Validation and Testing

To evaluate our Scooby models, we used an evaluation script developed by the Gagneur lab. This script loads a self-trained model checkpoint and generates pseudobulk predictions of gene or start site expression counts across different cell types, depending on what the model was trained to predict. The evaluation uses a held-out NeurIPS test set, which compares model predictions against true pseudobulk coverage counts. From here, a scatter plot of predicted vs. observed values is produced, along side a pair of Pearson correlation plots. Our primary evaluation metric is the average Pearson R correlation coefficient across all different genes.

4.4 Use of Generative AI

As a student with limited biology and genomics background, I utilized ChatGPT as a 'bio-consultant' through various portions of my writing. Through this process, generative AI was incredibly helpful in piecing together genomic thoughts and concepts. For any section of text where generative AI was consulted, I first wrote the given section in my own, initial thoughts, and then prompted ChatGPT to ensure what I had written made sense, asking it to aide in rephrasing anything that could be made clearer or represented more cohesively.

Chapter 5

Results

This chapter presents the results of our experiments, based on the methodology outlined in the previous chapter. The following sections describe each of our fine-tuned Scooby models' performance across various tasks, including RNA coverage prediction and transcription start site prediction.

5.1 RNA Coverage Prediction With 5' Data

The following section outlines the process and results of fine-tuning Scooby-5'-cov.

5.1.1 Training Overview

To evaluate the performance of Scooby-5'-cov, we used all ten batches of our 5' data, and re-inserted the human genome loci file defined in the original Scooby paper. This was done in order to confirm the model could meaningfully predict general RNA coverage while training on our 5' dataset.

5.1.2 Model Performance

The results of our fine-tuned Scooby model showed a weak, positive correlation across all loci and cells, which can be referenced in the scatter plot in figure 5.1. This shows predicted vs. actual counts for RNA expression coverage. We achieved a mean Pearson R correlation of **0.26** between predicted and observed gene expression levels across all cell types, shown in figure 5.3. For highly expressed genes, this correlation was much stronger, with a mean Pearson R of **0.77**, shown in figure 5.2.

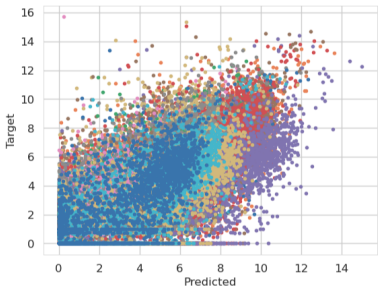

Figure 5.1: Scatter plot of Predictions vs. Targets

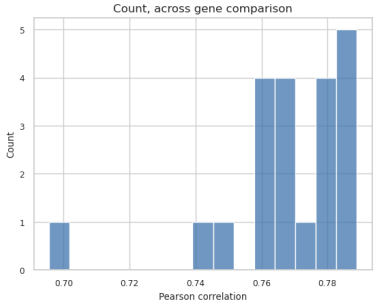

Figure 5.2: Histogram of Pearson R Scores for highly variable genes

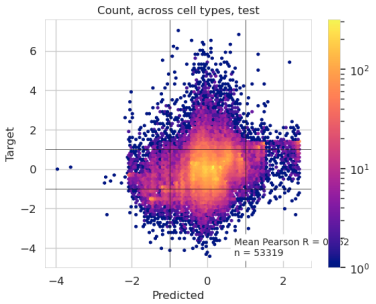

Figure 5.3: Cell Type-Level Predictions

Figure 5.4: Evaluation of Scooby-5'-cov Performance Across Metrics

5.1.3 Analysis

These results demonstrate that a model fine-tuned exclusively on 5' RNA-seq data can effectively generalize to broader RNA coverage prediction. While slightly lower than Scooby's original baseline, achieving a mean Pearson correlation of 0.26 (fig. 5.3) proves the model can learn meaningful transcriptional signal beyond the 5' end, despite being trained on a coverage-biased input. Even more impressive is the 0.77 Pearson R score (fig. 5.2) across highly expressed genes, as this matches the original Scooby-3'-cov model's performance. This suggests that 5' reads retain sufficient information on the structure of the transcript to support the modeling of full-length expression profiles. Scooby-5'-cov's performance highlights both the robustness of the fine-tuned Scooby architecture and the latent potential of

5' data as a foundation for transcriptome-wide prediction tasks. From here, we are able to advance into our primary goal of using 5' RNA-seq data to model TSS activity.

5.2 Transcription Start Site Prediction With 5' Data

The following section outlines the process and results of fine-tuning Scooby-5'-tss.

5.2.1 Training Overview

To test Scooby-5'-tss' performance on predicting transcription start sites, we initially fine-tuned the model using five of the ten batches from the fetal gut dataset, specifically those in the FCA-801 group, as described in Section 4.1.1. These batches provided a subset of cell types and coverage patterns for early experimentation. Following this initial training, we expanded the dataset to include all ten batches of data. This increased both the diversity and volume of the training data, in hopes of improving model accuracy and robustness. In full, Scooby-5'-tss predicts read counts across various TSS regions.

5.2.2 Model Performance

The initial results in TSS prediction from Scooby-5'-tss were strong. We see a weak, positive correlation between predicted and actual TSS coverage across all loci and cells in figure 5.5. Our model produced a mean Pearson R score of **0.23** across all cell types for TSS coverage prediction, seen in figure 5.7. Again, note the histogram showing Pearson R scores across genes is only for highly expressed/meaningful genes, or genes with an expression level greater than 1. We see from figure 5.6 the median Pearson R value of our fine-tuned model for these genes is **0.67**, showing a strong, positive correlation between predicted and observed values.

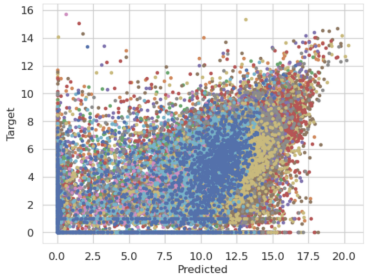

Figure 5.5: Scatter plot of Predictions vs. Targets

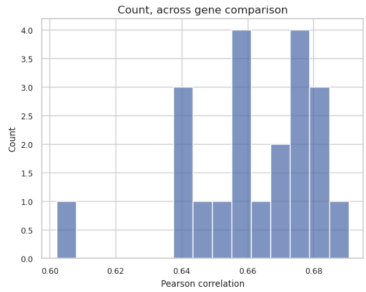

Figure 5.6: Histogram of Pearson R Scores for highly variable genes

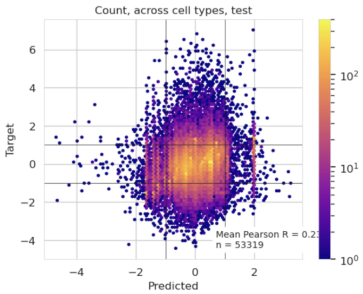

Figure 5.7: Cell Type-Level Predictions

Figure 5.8: Evaluation of Scooby-5'-tss (5-batch) Performance Across Metrics

After obtaining these strong results, we transitioned into a Scooby-5'-tss fine-tuning using all ten batches of gut data. First, the weak, positive correlation between predicted and actual TSS coverage across all loci and cells remained, but we seemed to have slightly less noisy predictions (see figure 5.9). With this, our model produced a **0.25** mean Pearson R score across cell types (shown in figure 5.11), only slightly outperforming our original fine-tuning with just five batches. On highly expressed genes, the performance dropped slightly to an R value of **0.66** (see figure 5.10).

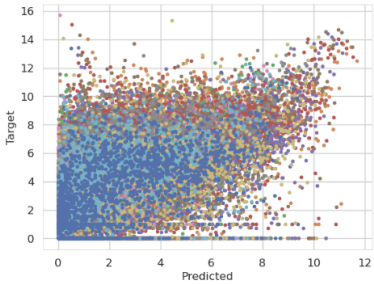

Figure 5.9: Scatter plot of Predictions vs. Targets

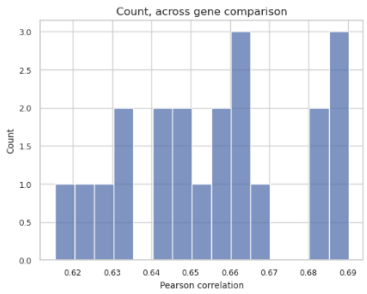

Figure 5.10: Histogram of Pearson R Scores for highly variable genes

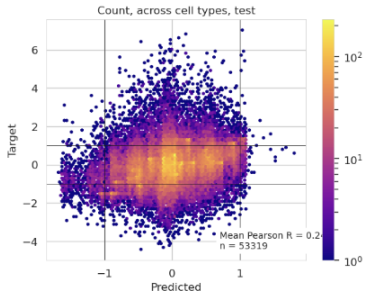

Figure 5.11: Cell Type-Level Predictions

Figure 5.12: Evaluation of Scooby-5'-tss (10-batch) Performance Across Metrics

5.2.3 Analysis

Scooby-5'-tss demonstrated strong, promising performance in predicting transcription start site coverage. We recorded an overall mean Pearson R correlation of 0.25 between predicted and observed TSS signal counts across all cell types (figure 5.11) for our final model fine-tuned on all ten batches of 5' gut data. While statistically weak, a meaningful signal was still captured by the model, despite the inherent complexity of the task. Notably, performance was considerably stronger when focused on highly expressed genes, with an average Pearson R correlation of 0.66 (figure 5.10). This suggests that Scooby-5'-tss' effectiveness lies in modeling transcriptional initiation within regions where biological signal is more consistent (just like Scooby-3'-cov). One thing to note is the limited increase in performance following the doubling of our training data. This points to the complexity of predicting TSS coverage and the notion that much more data could be utilized to increase performance.

Overall, the model's predictive ability on active TSS regions highlights its utility in leveraging 5' RNA-seq data for modeling transcriptional activity. The results broadly indicate an *expanded predictive scope* for Scooby and point towards promising avenues for further refinement. Together, these observations reinforce the value of integrating 5'-end transcriptomic data into predictive frameworks for RNA regulation.

5.2.4 Evaluating Model Performance With and Without Blood Cells

Our initial fine-tuning of Scooby was trained on all available 5' RNA-sequencing data from the FCA-801 and FCA-945 fetal gut datasets. At this point in our research, the dataset included a large number of blood cells, as we did not have access to organoid-specific cell type annotations yet. Once the cell type labels became available, it became clear that blood cells are not found in intestinal organoids, which is the biological system Scooby is designed to model. Since blood cells are transcriptionally distinct from solid tissue cell types, our fine-

tuned Scooby model could have learned to distinguish blood versus solid tissue more than it learned biologically relevant variation within solid tissue—particularly in early results. We hypothesized that this may have artificially boosted correlation scores, since distinguishing an abundant and separable population is easier than learning fine-grained promoter activity patterns. To test this, we repeated the fine-tuning on Scooby using only the initial five batches from the FCA-801 dataset, this time excluding blood cell types and retaining only: Epithelial, Mesenchymal, Endothelial, and Neuronal cells.

Model Results Without Blood Cells

After removing the blood cells from our initial five-batch fine-tuning dataset, we saw a **decreased** performance from Scooby-5'-tss. Figure 5.13 displays a weak, positive correlation between our predictions vs. actual values across loci and cells, however with less noisy predictions than those in figure 5.5. We achieved an average Pearson R score of **0.21** across cell types (figure 5.15) as opposed to our initial result of 0.24 (figure 5.7), which included the blood cells. Performance on highly expressed genes, however, remained constant with a R score of **0.66**, shown in figure 5.14.

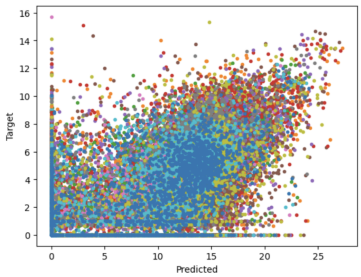

Figure 5.13: Scatter plot of Predictions vs. Targets

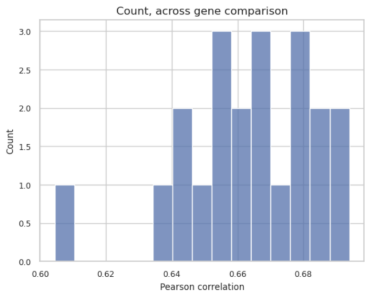

Figure 5.14: Histogram of Pearson R Scores for highly variable genes

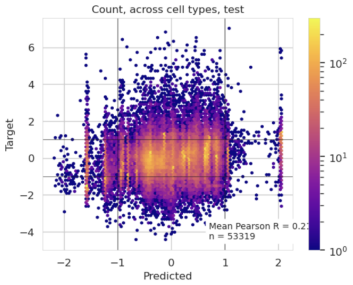

Figure 5.15: Cell Type-Level Predictions

Figure 5.16: Evaluation of Scooby-5'-tss (5-batch, No blood) Performance Across Metrics

When considering our core findings, the distinction between blood cells vs. non-blood cells in the training set is not detrimental, as our model still achieves promising results in

predicting TSS activity. However, this finding is important in showing how various factors in the makeup of your data can influence model performance, even if the effect is minimal.

5.2.5 BPNet and Other Model Considerations

Although it ultimately was outside of the scope of this thesis, we initially considered evaluating additional model types in using 5' data for TSS coverage prediction. Our main consideration was adapting the BPNet model[2], a convolutional neural network designed to predict transcription factor binding sites from DNA sequences. This model leverages high-resolution ChIP-nexus[1] data for validation, enabling precise predictions of both the distribution and strength of TF binding sites (ChIP-nexus data is a type of high-resolution experimental data that shows exactly where specific proteins, like transcription factors, bind to DNA). BPNet provides a higher level of detail compared to traditional methods through its modeling of binding patterns at single base resolution. A key advantage of BPNet is its ability to capture information from the entire DNA sequence, not just specific motifs, or sections of the DNA. The integration of sequence clustering and frequency scoring allows the model to provide a more holistic view of binding dynamics. This approach outperforms many traditional methods as it provides more accurate single-base-pair predictions. BPNet's use of modern machine learning techniques with high-resolution biological data represents a massive advancement in the field of computational genomics. With additional time and resources, we would have loved to explore BPNet's performance when fine-tuned using our 5' dataset. The goal would be to compare its performance to Scooby-5'-tss', helping us determine which model architecture (CNN or transformer) may be better suited for transcriptional activity prediction.

[1]See Appendix A.17 for at length definition

Chapter 6

Discussion

This section will summarize the thesis goals and discuss our accomplishments, shortcomings, and future considerations for the advancement of this work.

6.1 Summary of Thesis Goals

In this thesis, we set to adapt a previously trained transformer model, Scooby, to train on 5' RNA-seq data. In doing this, we first set out to train a model on this data to predict general RNA coverage, as the original Scooby model we adapted did. This would become Scooby-5'-cov. Following this checkpoint, we wanted to extend Scooby's capabilities, using the framework not only to train on 5' RNA-seq data, but to predict TSS coverage. This would become Scooby-5'-tss. In doing this, we hoped to expand the scope of the Scooby model, in turn uncovering a new direction in genomic machine learning modeling.

6.2 Conclusion

6.2.1 Scooby-5'-cov Performance

Scooby-5'-cov was successful, working as a launch-pad towards completing Scooby-5'-tss. It ultimately proved that 5' RNA-seq data *can* be used to predict general RNA expression data at single-cell resolution. While our earned Pearson R correlation of 0.26 between predicted and observed gene expression levels across all cell types fell short of Scooby-3'-cov's baseline R score of 0.32, the results confirm the feasibility of modeling gene expression coverage with 5' data. Further, it was even expected to see a decreased performance in prediction, as our 5' data is more start-site focused, so it will not generalize to expression coverage as well. A clear strong suit of Scooby-5'-cov was its ability to predict gene expression counts for highly expressed genes. It saw the same average R score (0.77) as Scooby-3'-cov did, again confirming that 5' RNA-seq is suitable for Scooby's original task.

6.2.2 Scooby-5'-tss Performance

Starting with areas for improvement, Scooby-5'-tss could benefit from increasing the number of cell batches used for fine-tuning, as well as decreasing the amount of noise in the data. We saw after removing blood cells from the data that our performance dropped from an R score of 0.26 to 0.21 on predicted vs. actual gene expression across cells–this tells us that the patterns Scooby-5'-tss learned may have been slightly less meaningful than our surface level conclusions suggest (specifically on our initial training with 5 batches). With more, non-blood cell data, we could potentially improve our results.

Overall, Scooby-5'-tss showed great success. We achieved an average Pearson R score of 0.25 for predicted vs. observed TSS counts across all cells; while statistically weak, this was a massive result in confirming the plausibility in modeling TSS activity with 5' data. The performance was slightly below our Scooby-3'-cov baseline performance, which predicted gene expression at an R level of 0.32 across all cell types. In addition Scooby-5'-tss showed

strong results for TSS prediction on highly expressed gene groups, with a Pearson R score of 0.67 (again slightly lower than our Scooby-3'-cov reference performance of 0.77). While we did not outperform the original Scooby model (in performance numbers), our success in adapting it to predict *a new coverage type* is evident. With these results, we can now confirm that it is possible to model transcription start site coverage using 5' data in a transformer architecture. Through this success, we have also expanded the scope and capabilities of Scooby, fulfilling our overarching goal.

6.3 Future Considerations

6.3.1 Data Improvement

As addressed in the previous section, one area for improvement in our models would be increasing the quantity of our data. We used 57k cells in our fine tuning of Scooby-5'-tss, and while it is unknown the number of cells the Gagneur lab leveraged in the original Scooby[1] paper, more is always better when it comes to machine learning training sets. In doing so, we would hope to train with a larger set of non-blood cells, in hopes of increasing meaningful performance through the reduction of noisy data.

6.3.2 Use of Convolutional Neural Networks

As noted in section 5.2.5, we originally hoped to adapt a model using BPNet[2], a CNN model designed for predicting transcription factor binding sites from DNA sequences. While it ultimately fell out of our immediate scope, it is among our top considerations for future work and improvements. As seen in our own use of Scooby, transformer models typically feature down and upsampling techniques, like pooling or strided convolutions. In doing this, the loss of fine-grained spacial precision can appear, especially in short genomic contexts like transcription start sites. Fully convolutional neural networks like BPNet can process their inputs at a base-pair resolution, meaning they can avoid down or upsampling. In this,

they have the ability to better preserve patterns in fine-scale motifs. In knowing this, a model with a CNN architecture like BPNet may be better suited for high performance in TSS coverage prediction, when paired with the more fine-grained sequences in 5' data. With that said, our next consideration following this paper would be to attempt to achieve higher prediction performance through this methodology, now that we have proved TSS coverage prediction is possible.

Appendix A

Definitions of Key Terms

A.1 — RNA Sequencing: RNA Sequencing is a scientific process that allows researchers to reveal the quantity of RNA molecules in a cell sample, which gives insight into the given cell's gene expression.

A.2 — 3' RNA Sequencing: 3' RNA sequencing is a technique which sequences the 3' ends of polyadenylated RNA transcripts. While cost-effective and scalable, it lacks information on transcript start sites, isoforms, and alternative splicing events.

A.3 — 5' RNA Sequencing: 5' RNA sequencing sequences the 5' ends of RNA transcripts, providing information about transcription start sites and promoter usage. This method enables the detection of cell-type-specific TSS activity and alternative initiation events. Unlike 3' RNA-seq, it offers higher resolution of gene regulation at the level of transcription initiation.

A.4 — ATAC Sequencing: ATAC Sequencing is a scientific process that allows researchers to reveal accessible sections of the DNA. Once performed, results can be used to interpret transciption factor binding sites and nucleosome positions.

A.5 — Lineage Tracing: Lineage tracing is a scientific process used to track the life cycles of individual or groups of cells. Using CRISPR-Cas9 technology, barcodes are inserted into cells' genomes as a way to 'tag' them–from here, scientists can perform various

experiments on these cells and watch them develop.

A.6 — Self-Attention: Self-attention is a mechanism that allows neural networks to weigh the importance of different input elements while making predictions. This allows models to capture long-range dependencies effectively. It computes attention scores between all elements in an input sequence, generating a weighted sum of values that emphasize the most relevant features. Self-attention is a core component of transformer models, which have revolutionized the field of machine learning.

A.7 — Enformer Model: The Enformer model is a deep learning architecture developed to predict gene expression and regulatory activity directly from the raw genomic DNA sequence. It uses transformer layers to model long-range interactions across the genome, producing accurate predictions of genomic outputs such as CAGE-seq, ATAC-seq, and histone modifications.

A.8 — U-Net Connections: U-Net is a convolutional neural network architecture originally designed for image segmentation. It uses skip connections to combine high-resolution spatial features from earlier layers with deep semantic information from later layers, in turn allowing for precise localization and context-aware predictions.

A.9 — Low-Rank Adaptation: Low-Rank Adaptation, or LoRA, is a machine learning method to fine-tune large ML models. In LoRA, a majority of the weights in a transformer model are frozen, and separately fine-tuned weights are added to each layer. This greatly reduces the computational cost and time of training large models, and is an incredibly effective method of fine-tuning.

A.10 — SnapATAC (Python package): SnapATAC is a Python module which allows programmers to efficiently analyze single-cell ATAC-seq datasets.

A.11 — Peft (Python package): Peft, which stands for *Parameter-Efficient Fine-Tuning*, is a HuggingFace module that allows programmers to apply LoRA to transformer models.

A.12 —JASPAR: JASPAR is an open access database of curated, non-redundant transcription factor binding profiles stored as position frequency matrices.

A.12 — Principle Component Analysis: PCA is a linear dimensionality reduction technique. It is often utilized for exploratory data analysis, and allows researchers to break down complex data components to be smaller and more meaningful.

A.14 — Uniform Manifold Approximation and Projection: UMAP is a non-linear dimensionality reduction technique. It has similar applications as the aforementioned PCA.

A.15 — Leiden Clustering: Leiden clustering is a community detection algorithm typically used for clustering single-cell RNA-sequencing data.

A.16 — Wilcoxon Signed-Rank Test: The Wilcoxon signed rank test is used for statistical hypothesis testing, to test the location of a population based on a sample of data, or to compare the locations of two populations.

Appendix B

Scooby Architecture

Input. As input to Scooby, the model takes the 5' single-cell RNA-seq data in as an AnnData object (which holds cell-level metadata), and as fragment files for per-read data. Per training example, we provide the model with an input DNA sequence 524 kb long, as a one-hot encoded sequence (A = [1,0,0,0], C = [0,1,0,0], T = [0,0,1,0], and G = [0,0,0,1]), and an output RNA signal in the form of a vector of binned read counts (with separate tracks for RNA+ and RNA- strands). Training targets were generated each iteration using a dataset-specific class that dynamically sampled coverages from both + and - strand fragment-backed AnnData objects. For each locus, the dataset sampled 64 neighboring cells in a precomputed cell to cell similarity graph, which enabled the model to learn from smoothed pseudo-bulk profiles that reflect local cell variation. Genomic loci were filtered based on length and distance from chromosome ends to ensure a valid context window for every sample.

Model Architecture. The transformer backbone is initialized from the pre-trained Borzoi model used in the original Scooby paper (johahi/borzoi-replicate-0), which includes eight transformer layers, each composed of a multi-head attention block (with rotary positional encodings) followed by a multi-layer perceptron. Prior to the transformer blocks, the sequences pass through various convolutional and max-pooling layers, which breaks the input to 4,096 embeddings at a 128 base-pair resolution. Each embedding has a dimension-

ality of 50, which was computed through PCA. From here, a cell-specific decoder is utilized, which maps the transformer output to predicted RNA signals. This decoder output is then upsampled to a resolution of 32 bp per bin, producing a final output of 196,608 bp over two RNA strands.

Convolutional Smoothing of RNA Targets

To reduce sparsity in the RNA coverage signal, we applied a smoothing and scaling operation to our target signals. Let $y = (y_1, y_2, \ldots, y_L)$ denote the original per-base coverage vector. The smoothed and downsampled signal $\tilde{y}_m$ at bin m was computed as:

$$\tilde{y}_m = \underbrace{c_{\text{upscale}}}_{\text{ours}} \cdot \left(\underbrace{k}_{\text{default}} \cdot \frac{1}{k} \sum_{i=1}^{k} y_{(m \cdot s)+i-\lfloor k/2 \rfloor} \right)$$

$$= c_{\text{upscale}} \cdot \sum_{i=1}^{k} \frac{1}{k} \cdot y_{(m \cdot s)+i-\lfloor k/2 \rfloor}$$

Where:

$k = 32$ (kernel size)

$s = 32$ (stride)

$c_{\text{upscale}} = 20$ (scaling factor applied after pooling)

L is the length of the original signal vector

This equation represents a fixed average over non-overlapping 32-base windows (with stride 32), followed by two stages of implemented scaling: First, a fixed factor of 32 to reverse the averaging operation (used in the original Scooby implementation), and second, an additional up-scaling of 20 to amplify smoothed signal intensity in 5' RNA-seq data. Thus, the total effective scaling coefficient is 640.

Target Normalization

In another effort to address the numerical instability of our target values (most of which

were very close to 0), we applied a z-score normalization to our target vectors before our loss step. In doing this, we eliminated issues with poor gradient flow and saw increased model performance. For each set of targets, the normalized signal $\tilde{y}_i$ at bin i was computed as:

$$\tilde{y}_i = \frac{y_i - \mu}{\sigma}$$

Where:

y_i is the original target value at bin i

μ is the mean of the target vector across all bins

σ is the standard deviation across bins

Loss Function. For fine-tuning, we utilized a poisson multinomial loss function, designed for modeling count-based sequencing data. It models our predicted output as a multinomial distribution over binned read counts, where the total count follows a Poisson distribution. This helps capture both local coverage patterns (through the multinomial shape) and global expression levels (through the Poisson total). To achieve our best results (and reproduce the original Scooby results), we removed sequence-length normalization from the multinomial and poisson terms in the loss function. This step is equivalent to introducing coefficients w_{mult} and w_{pois}, and decreasing their values from L to 1. This process is highlighted by the following equation:

$$\mathcal{L}_{\text{PM}} = \frac{1}{B} \sum_{b=1}^{B} \left[w_{\text{mult}} \cdot \sum_{i=1}^{L} -y_{b,i} \log \left(\frac{\hat{y}_{b,i}}{\sum_j \hat{y}_{b,j}} \right) + w_{\text{pois}} \cdot \left(\hat{N}_b - N_b \log \hat{N}_b \right) \right]$$

where: B is the batch size

L is the number of bins (sequence positions)

$y_{b,i}$ is the observed count in bin i for sample b

$\hat{y}_{b,i}$ is the predicted (non-normalized) value in bin i for sample b

$N_b = \sum_i y_{b,i}$ is the total observed count for sample b

$\hat{N}_b = \sum_i \hat{y}_{b,i}$ is the total predicted count for sample b

w_{mult} is the weight for the multinomial term

w_{pois} is the weight for the Poisson term

Fine-tuning. We fine-tuned for 40 epochs using the AdamW optimizer, with a batch size of one. We evaluated the model every 256 training steps, using 'chromosome 14' as a held-out validation set. While we trained for 40 epochs, our best recorded performance for both Scooby-5'-cov and Scooby-5'-tss were between epochs five and ten. In fine-tuning, LoRA was used to fine-tune only a subset of transformer weights related to RNA prediction. During this training, a shift augmentation was applied to each sequence, which randomly shifts the input DNA sequence window left or right (+/-3 bins). This helped the model learn position-invariant features and improved generalization. Gradient clipping was also used to limit the magnitude of gradients during backpropagation. This was done to prevent exploding gradients, which would destabilize training. More specifically, we clipped gradients to a maximum global norm of one. Finally, we utilized a learning rate of 1e-5 and a weight decay of 1e-4.

Bibliography

[1] J. C. Hingerl, L. D. Martens, A. Karollus, T. Manz, J. D. Buenrostro, F. J. Theis, and J. Gagneur, "scooby: Modeling multi-modal genomic profiles from DNA sequence at single-cell resolution," Sept. 2024.

[2] J. Linder, D. Srivastava, H. Yuan, V. Agarwal, and D. R. Kelley, "Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation," *Nature Genetics*, pp. 1–13, Jan. 2025.

[3] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-r. Loh, and S. Raychaudhuri, "Fast, sensitive and accurate integration of single-cell data with Harmony," *Nature Methods*, vol. 16, pp. 1289–1296, Dec. 2019.

[4] "gagneurlab/scooby_reproducibility," Mar. 2025. original-date: 2024-09-18T12:05:10Z.

[5] "gagneurlab/scooby," Mar. 2025. original-date: 2024-09-18T11:29:53Z.

[6] Avsec, M. Weilert, A. Shrikumar, S. Krueger, A. Alexandari, K. Dalal, R. Fropf, C. McAnany, J. Gagneur, A. Kundaje, and J. Zeitlinger, "Base-resolution models of transcription-factor binding reveal soft motif syntax," *Nature Genetics*, vol. 53, pp. 354–366, Mar. 2021.

[7] R. Andersson and A. Sandelin, "Determinants of enhancer and promoter activities of regulatory elements," *Nature Reviews Genetics*, vol. 21, pp. 71–87, Feb. 2020.

[8] R. Andersson, C. Gebhard, I. Miguel-Escalada, I. Hoof, J. Bornholdt, M. Boyd, Y. Chen, X. Zhao, C. Schmidl, T. Suzuki, E. Ntini, E. Arner, E. Valen, K. Li, L. Schwarzfischer, D. Glatz, J. Raithel, B. Lilje, N. Rapin, F. O. Bagger, M. Jørgensen, P. R. Andersen, N. Bertin, O. Rackham, A. M. Burroughs, J. K. Baillie, Y. Ishizu, Y. Shimizu, E. Furuhata, S. Maeda, Y. Negishi, C. J. Mungall, T. F. Meehan, T. Lassmann, M. Itoh, H. Kawaji, N. Kondo, J. Kawai, A. Lennartsson, C. O. Daub, P. Heutink, D. A. Hume, T. H. Jensen, H. Suzuki, Y. Hayashizaki, F. Müller, A. R. R. Forrest, P. Carninci, M. Rehli, and A. Sandelin, "An atlas of active enhancers across human cell types and tissues," *Nature*, vol. 507, pp. 455–461, Mar. 2014.

[9] D. E. Wagner and A. M. Klein, "Lineage tracing meets single-cell omics: opportunities and challenges," *Nature Reviews Genetics*, vol. 21, pp. 410–427, July 2020.

[10] "Transformer (deep learning architecture)," Mar. 2025. Page Version ID: 1282085317.

[11] J. Hingerl and L. D. Martens, "scooby: Modeling multi-modal genomic profiles from DNA sequence at single-cell resolution - Supplementary and training data," Oct. 2024.

[12] Avsec, V. Agarwal, D. Visentin, J. R. Ledsam, A. Grabska-Barwinska, K. R. Taylor, Y. Assael, J. Jumper, P. Kohli, and D. R. Kelley, "Effective gene expression prediction from sequence by integrating long-range interactions," *Nature Methods*, vol. 18, pp. 1196–1203, Oct. 2021.

[13] "Wilcoxon signed-rank test," Nov. 2024. Page Version ID: 1259617537.

[14] "JASPAR: An open-access database of transcription factor binding profiles."

[15] "johahi/borzoi-replicate-0."

[16] "ChatGPT."